Referee Report on *"Reddit Users Unleashed – Understanding User Behaviour and Their Impact on Meme Stocks"*

Summary of the Paper

This paper studies the relationship between social network activity on Reddit and stock market dynamics during the GameStop (GME) market frenzy. The authors construct a large communication network from posts and comments on the *WallStreetBets* (WSB) subreddit, covering June 2020 to May 2021 and involving more than 500,000 users. The objective is to understand whether changes in users' behaviour in the social network help explain stock market movements during the meme stock episode.

Assessment

The paper studies an interesting and timely topic, and the dataset used in the paper is quite unique and offers the possibility to study large-scale social network interactions. However, several aspects of the analysis would benefit from further clarification and development. In particular, the econometric contribution could be articulated more clearly, the identification strategy strengthened, and the economic mechanisms underlying the results discussed in greater detail.

Major Comments

1. *Contribution relative to the econometric literature.* The paper proposes a new econometric framework (ISBM), but the novelty of the methodological contribution relative to existing approaches could be articulated more clearly. The framework combines several components—logistic regression models for link prediction in networks, change-point detection methods, and autoregressive models linking network variables to financial outcomes. It would therefore be helpful if the authors clarified more explicitly what the main methodological contribution of the ISBM framework is, how it differs from existing dynamic network models, and whether the proposed change-point procedure introduces any new econometric challenges. More generally, given the journal's focus, the paper could benefit from a more rigorous and

compact presentation of the econometric methodology, ideally in a more general form that could be adapted to different empirical applications beyond the specific setting studied in the paper.

2. *Identification and causal interpretation.* The distinction between influential and non-influential users appears somewhat arbitrary, and it would be helpful to see additional robustness checks to assess how sensitive the results are to this classification (beyond 10% thresholding). In particular, prominent users (such as Keith Gill) may generate substantial attention within the network alone, and should be controlled for/ be partialed out.

3. *Interpretation of the main empirical findings.* One of the most intriguing findings of the paper is that behavioral changes among less influential users account for a larger share of market dynamics than those among influential users. While this result is interesting, the paper could benefit from a deeper discussion of the mechanisms behind it. At the same time, the results suggest a timing gap between influencers and the broader user base: influential users appear to change their behavior earlier—sometimes weeks before the market frenzy—while the crowd's collective behavior is associated with more immediate market reactions. This raises the question of whether influencers and followers operate on different information sets or whether information diffuses through the network over time (spillover effects).

4. *Model scalability, robustness and sparsity.* While the authors emphasize the computational feasibility of the approach, the discussion of the underlying assumptions that make the estimation tractable remains limited. (i) The sparsity assumptions imposed on the network links appear to play a key role in reducing the dimensionality of the problem, yet it is not entirely clear how sensitive the results are to these assumptions or whether alternative sparsity structures would lead to similar conclusions. (ii) The number of parameters increases rapidly once the change-point specification is introduced, as it requires estimating multiple parameter matrices across regimes. The objective function for estimating these is not explicitly stated. A clearer discussion of the computational complexity and the role of sparsity—possibly including robustness checks—would help clarify the scope of the methodology and guide practitioners interested in applying the framework in other large-dimensional network settings.

Data Construction

Several aspects of the data construction require clarification:

- The data description should state the source of intraday GME price data.
- How are multiple comments occurring within the same hour handled?
- How are comments in the same thread encoded in the directed network structure?